EXP NO: 4 DATE: 24.02.2026

VERIFICATION OF THEVENIN'S THEOREM:

AIM:

To verify Thevenin's theorem practically and theoritically for the given DC circuit.

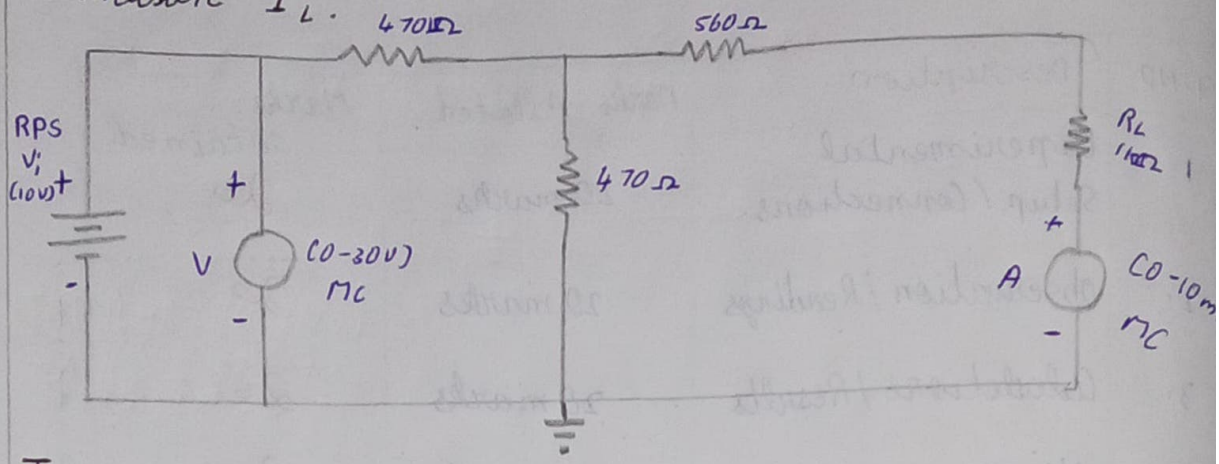
APPARATUS REQUIRED:

SL.NO	APPARATUS	SPECIFICATION	QUANTITY
1	Regulated Power Supply (RPS)	(0-30V)	1
2.	Voltmeter	(0-30V) MC	1
3.	Ammeter	(0-10mA) MC	1
4.	Resistors	470 Ω 560 Ω 1 KΩ	2 1 1
5.	Bread board	--	1
6.	Multimeter	--	1

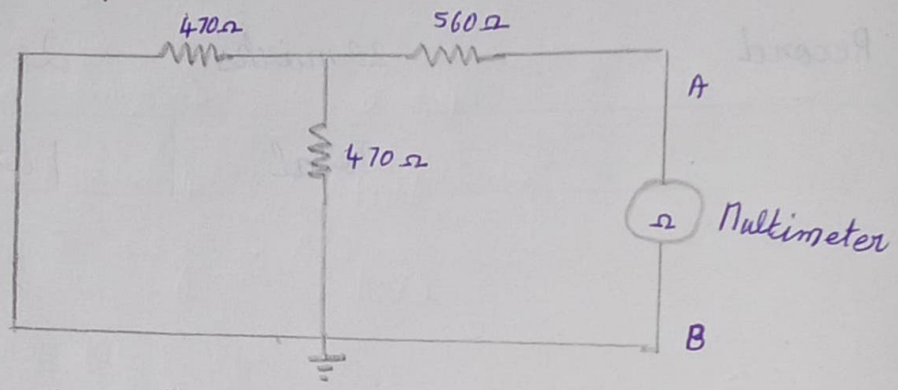
CIRCUIT DIAGRAM:

VERIFICATION OF THEVENIN'S THEOREM:

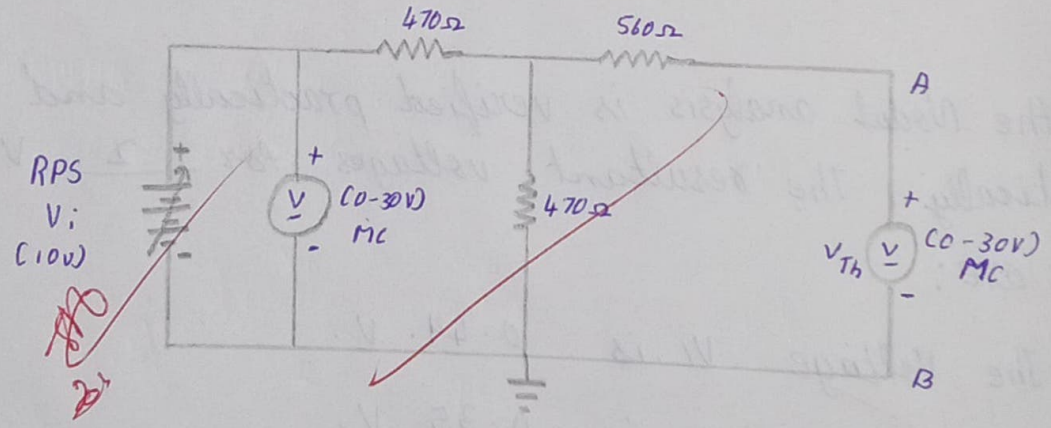
To measure I_L.



To measure R_{Th} or R_N:



To measure V_{Th} or V_{oc}:



PROCEDURE:

1) Make the connections as per the Circuit Diagram:)

2) Vary the RPS and set an input voltage of 10V.

3) Note down the voltmeter reading (V_i) and ammeter reading (I_L) in Tabular Column 1.

4) Switch off the supply and make connections for Circuit Diagram 2.

5) Measure the Thevenin's resistance R_{Th} = Norton's resistance R_N.

6) Switch off the supply and make connections for circuit diagram 3.

7) Set an input voltage of 10V in the RPS and note down the voltmeter reading V_i and Ammeter reading I_N ($= I_{sc}$) in Tabular Column 4.

8) Draw the Thevenin's equivalent circuit and Norton's equivalent circuit as shown in circuit diagram 5 & 6 respectively.

9) Calculate the I_L value using the formula

Thevenin's Theorem

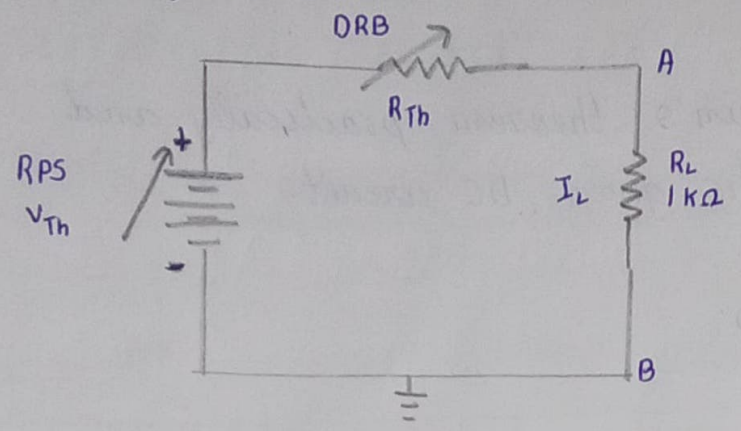
$$I_L = V_{Th} / (R_{Th} + R_L)$$

Norton's Theorem

$$I_L = I_N * R_N / (R_N + R_L)$$

10) Theoritically verify the Norton's theorem.

Thevenin's equivalent circuit:



TABULAR COLUMN : 1

To measure I_L

V_i (Volts)	I_L (amps)
2	0.59
4	1.16
6	1.73

TABULAR COLUMN : 2

To measure R_{Th} or R_N

From circuit diagram 2, $R_{Th} = R_N = \underline{722}\ \Omega$

TABULAR COLUMN : 3

To measure V_{Th} or V_{oc}

V_i (Volts)	V_{Th} (Volts)
2	1.06
4	2.09
6	3.10

OK verified
21/2/22

MODEL CALCULATION :

Practical value of I_L (from tabulation 1) = 2.3mA

Verification of Thevenin's theorem:

$$I_L = V_{Th} / (R_{TH} + R_L) = 2.22mA$$

Theoretical calculation of I_L, R_{TH} (R_N) and V_{Th} for the given circuit:

$$\Rightarrow R_R = \frac{470 \times 470}{470+470} = \underline{235\Omega}$$

$$\Rightarrow V_{th} = \frac{V_i \times R_R}{R_L + R_p} = \frac{2 \times \cancel{470}\,235}{470+235} = 0.66V$$

For Calculating R_{TH}:

470 parallel to 470 = 235 Ω

R_{TH} = 560 + 235 = 795 Ω

For Calculating I_L;

$I_L = V_{TH} / R_{TH} + R_L$

= 323 / 795 + 1000

= 1.88mA

MARKS SPLIT-UP:

Sl. No	Description	Marks Allotted	Marks obtained
1	Experimental Setup/ Connections	20 marks	20
2.	observation / Readings	20 marks	20
3.	Calculations / Results	20 marks	20
4.	Viva	20 marks	20
5.	Record	20 marks	20
		Total	100

100

25/2/26

RESULT:

Thus Thevenin's theorem is verified practically and theoretically.